AI FUNDAMENTALS

A Comprehensive Introduction to Artificial Intelligence

Proxiant Academy

4-Week Intensive Program

COURSE INFORMATION

Course Title	AI Fundamentals: A Comprehensive Introduction to Artificial Intelligence
Institution	Proxiant Academy
Duration	4 Weeks (Intensive)
Format	Blended Learning (Virtual + Asynchronous Materials)
Prerequisites	None. Basic Python familiarity and high-school mathematics are recommended but not required.
Instructor	Pavan R
Office Hours	Twice weekly; schedule announced at cohort start
Course Level	Beginner to Intermediate

COURSE DESCRIPTION

AI Fundamentals is a comprehensive 4-week intensive course designed to provide participants with a solid understanding of artificial intelligence concepts, techniques, and applications. This course bridges the gap between theoretical foundations and practical implementations, making AI accessible to both technical and non-technical audiences. Drawing from leading university curricula including MIT 6.034, Stanford CS221, and Carnegie Mellon's AI programs, this course covers the essential building blocks of modern artificial intelligence.

Participants will explore the evolution of AI from early symbolic systems to contemporary deep learning approaches. The curriculum emphasizes hands-on learning through weekly assignments and projects, ensuring that students not only understand the theory behind AI systems but can also build and implement them. Upon successful completion, students become eligible for the Proxiant Certified AI Professional (PCAP) certification, which demonstrates mastery of AI fundamentals. By the end of this course, participants will have the knowledge and practical skills to understand AI systems, evaluate their capabilities and limitations, and apply AI techniques to real-world problems.

This course is ideal for professionals seeking to understand AI's impact on their industry, software developers looking to incorporate AI into their applications, business leaders evaluating AI solutions, and anyone curious about the future of artificial intelligence. No prior experience with AI or machine learning is required, though basic programming knowledge and comfort with mathematics is recommended.

LEARNING OUTCOMES

Upon successful completion of this course, participants will be able to:

1. Define artificial intelligence and understand its history, current state, and future directions in both academic and industry contexts
2. Distinguish between artificial intelligence, machine learning, and deep learning, understanding their relationships and differences
3. Formulate problems as search spaces and apply fundamental search algorithms (BFS, DFS, A*) to find solutions
4. Understand and implement supervised learning algorithms including regression, classification, decision trees, and support vector machines
5. Apply unsupervised learning techniques for clustering and dimensionality reduction to extract patterns from unlabeled data
6. Build and train neural networks from first principles, understanding forward propagation, backpropagation, and optimization techniques
7. Apply specialized deep learning architectures (CNNs, RNNs, Transformers) to different types of data and problems
8. Understand Large Language Models and generative AI, including prompt engineering and responsible use of foundation models
9. Analyze ethical implications of AI systems including bias, fairness, safety, and privacy considerations
10. Evaluate AI solutions for business applications and develop strategies for AI adoption in organizational contexts

COURSE SCHEDULE

The course is organized into four intensive weeks, each containing three sessions. Each week builds on the prior, progressing from AI foundations through machine learning, deep learning, and into generative AI and strategy.

WEEK 1: FOUNDATIONS OF ARTIFICIAL INTELLIGENCE			
#	**Session**	**Key Topics**	**Deliverables**
1	What is AI?	History, definitions, Turing Test, types of AI (Narrow / General / Super), rational agents	Week 1 exercise set
2	AI Paradigms	AI vs ML vs DL, intelligent agents, PEAS framework, environments, problem-solving as search	Week 1 exercise set
3	Search & Knowledge	BFS, DFS, A* search, heuristics, propositional logic, first-order logic, knowledge representation	Assignment 1 due Project 1 assigned

Readings: *Russell & Norvig Ch. 1–4; Historical Overview of AI*

WEEK 2: MACHINE LEARNING FUNDAMENTALS			
#	**Session**	**Key Topics**	**Deliverables**
4	Supervised Learning I	ML pipeline, linear regression, logistic regression, gradient descent, loss functions	Week 2 exercise set
5	Supervised Learning II	Decision trees, random forests, SVMs, ensemble methods, regularization	Week 2 exercise set
6	Unsupervised	K-means clustering, PCA, cross-validation, bias-	Assignment 2 due Project

	& Evaluation	variance tradeoff, precision / recall / F1	2 assigned

Readings: *Stanford CS221 lecture notes; Andrew Ng's ML materials; Intro to Statistical Learning Ch. 2–9*

WEEK 3: DEEP LEARNING AND NEURAL NETWORKS			
#	**Session**	**Key Topics**	**Deliverables**
7	Neural Networks	Perceptrons, activation functions, backpropagation, SGD, multi-layer networks, regularization	Week 3 exercise set
8	CNNs & RNNs	Convolutional layers, pooling, image classification; RNNs, LSTMs, sequence modeling	Week 3 exercise set
9	Transformers & NLP	Self-attention, multi-head attention, encoder-decoder, transfer learning, word embeddings, NLP basics	Assignment 3 due Project 3 assigned

Readings: *Goodfellow et al. Ch. 5–10; "Attention Is All You Need" (Vaswani et al., 2017); Stanford CS224N materials*

WEEK 4: GENERATIVE AI, ETHICS, AND AI STRATEGY			
#	**Session**	**Key Topics**	**Deliverables**
10	Generative AI & LLMs	LLMs, scaling laws, GPT architecture, RLHF, prompt engineering, RAG, AI agents	Week 4 exercise set
11	Ethics & Regulation	Bias and fairness, explainability (LIME/SHAP), AI safety, privacy, EU AI Act, NIST RMF	Week 4 exercise set
12	Industry & Strategy	AI across industries, organizational readiness, AI roadmaps, future trends, PCAP prep, course review	Assignment 4 due Project 4 due Final Exam

Readings: *Recent LLM papers; AI Ethics guidelines; Industry case studies; McKinsey / Gartner AI reports*

ASSESSMENT STRUCTURE AND GRADING

Your grade in AI Fundamentals is determined by the following components:

Assessment Component	Weight
Weekly Exercises	15%
Weekly Assignments	20%
Weekly Projects	30%
Final Certification Exam	35%
PASSING REQUIREMENT	70% Overall (PCAP Certification Eligible)

All assessments are graded on a 0-100 scale. The final course grade is calculated as a weighted average of all assessment components. A passing grade of 70% is required to receive the AI Fundamentals Certificate of Completion and become eligible for the Proxiant Certified AI Professional (PCAP) exam.

PCAP CERTIFICATION

Upon successful completion of AI Fundamentals with a passing grade of 70% or higher, you become eligible to take the Proxiant Certified AI Professional (PCAP) exam. The PCAP is an industry-recognized credential that validates your expertise in artificial intelligence fundamentals.

The PCAP exam consists of 100 multiple-choice questions to be completed in 3 hours. A score of 70% (70 out of 100 questions) is required to pass and earn the credential. The PCAP certification is valid for 3 years from the date of certification. The exam fee is $199. Upon successful completion, you will receive a digital badge and a verifiable online credential that can be shared with employers and added to your professional profiles.

TENTATIVE SESSION SCHEDULE

The following schedule is tentative and subject to change at the instructor's discretion:

Week	Session	Topic	Deliverables
Week 1: Foundations	1	What is AI? History, Definitions, Turing Test, Types of AI (Narrow/General/Super)	Week 1 exercise set
	2	AI vs ML vs DL, Intelligent Agents, Problem Solving as Search	Week 1 exercise set
	3	Search Algorithms (BFS, DFS, A*), Knowledge Representation, Logic	Assignment 1, Project 1 Assigned
Week 2: ML Fundamentals	4	ML Pipeline, Supervised Learning, Linear/Logistic Regression, Gradient Descent	Week 2 exercise set
	5	Decision Trees, Random Forests, SVMs, Ensemble Methods	Week 2 exercise set
	6	Unsupervised Learning, Model Evaluation, Bias-Variance Tradeoff	Assignment 2, Project 2 Assigned
Week 3: Deep Learning	7	Neural Networks Fundamentals, Perceptrons, Activation Functions, Backpropagation	Week 3 exercise set
	8	CNNs for Computer Vision, RNNs for Sequential Data, LSTMs	Week 3 exercise set
	9	Transformers, Attention Mechanism, Transfer Learning, NLP Fundamentals	Assignment 3, Project 3 Assigned
Week 4: GenAI & Ethics	10	Large Language Models, Generative AI, Prompt Engineering	Week 4 exercise set
	11	AI Ethics, Bias, Fairness, Safety, Regulatory Landscape	Week 4 exercise set
	12	AI in Industry, AI Strategy, Future of AI, Course Review	Assignment 4, Project 4, Final Exam

Dates will be confirmed at the start of the course. All times are in the instructor's local time zone unless otherwise specified.

REQUIRED AND RECOMMENDED RESOURCES

Required Textbooks

- Russell, S. J., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.

Recommended Resources

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning (2nd ed.). Springer.
- Stanford CS221: Artificial Intelligence - Lecture notes and materials available online
- MIT 6.034: Artificial Intelligence - Lecture videos and course materials
- Python libraries: NumPy, Pandas, Scikit-learn, TensorFlow/PyTorch, Jupyter Notebooks
- Recent papers on Large Language Models, Transformers, and AI Ethics (provided via course portal)

COURSE POLICIES AND EXPECTATIONS

Attendance and Participation

Attendance at all live sessions is strongly encouraged. Participants who miss sessions are responsible for obtaining notes from classmates and reviewing recorded materials. Regular participation in exercises and discussions contributes to the learning environment and your overall understanding of the material.

Late Submission Policy

Assignments and projects are due on the dates specified in the course schedule. Late submissions will be accepted up to 48 hours after the deadline with a 10% penalty per day. After 48 hours, submissions will not be accepted unless prior permission has been obtained from the instructor. Extensions may be granted for documented emergencies or health issues.

Academic Integrity

All coursework must be your own original work. While collaboration is encouraged for understanding concepts, all submitted assignments and projects must reflect your individual understanding and effort. Proper attribution must be given for any code, ideas, or resources that are not your own. Plagiarism and academic dishonesty will result in course failure and may lead to further disciplinary action.

Communication

The primary communication channel will be the course discussion forum. Email inquiries will be responded to within 24 business hours. Office hours are held twice weekly (schedule announced at cohort start) for one-on-one discussions. For technical issues, contact the course support team.

Accessibility and Accommodations

Proxiant Academy is committed to providing equal access to education. If you have a documented disability or require accommodations, please contact the accessibility office at the start of the course. We will work with you to provide appropriate accommodations to ensure your success in the program.

Technical Requirements

You will need a computer with Python 3.8+ installed, a reliable internet connection, and access to the course learning management system (LMS). All software tools used in this course are free and open-source. Technical support will be provided for course-specific platforms only.